

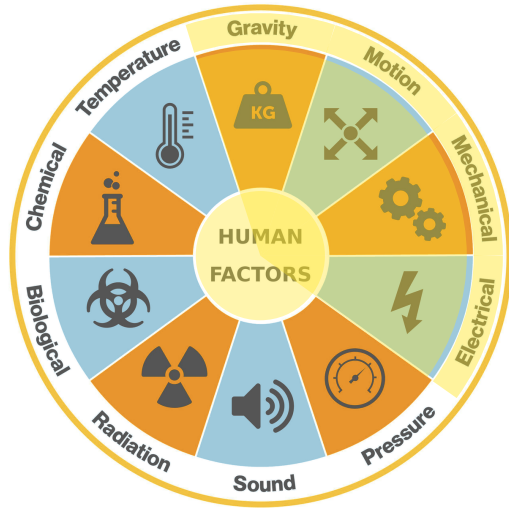
Activity Hazard Analysis

Location: I-40 / Business 40 utility relocation - Sta. 114+50 to 128+00, EB shoulder near Exit 285, Raleigh, NC	
Date (dd-mm-yyyy): 14-08-2026	JHA / procedure numbers: JHA-2026-0147, SOP-114, ITS-EXC-09
Person in charge: Miguel Rodriguez	Emergency number: 911 / Site safety: (919) 555-0182
Closest emergency centre: WakeMed Raleigh Campus - 3000 New Bern Ave	Is a rescue plan required for the work being performed? ☒ Yes ☐ No
Work order / permit number: WO-88213 / Permit E-4471	Muster point: North parking lot, gate 3
Description of work performed on site and activities happening in the vicinity of work area: Excavation and directional bore for fiber conduit relocation along EB shoulder; potholing existing utilities; loading and hauling spoils. Adjacent lane closures with live traffic; paving crew working 200 ft east of Sta 126 (simultaneous operations).	

Task	Hazards	Controls
Excavation around existing utilities	Cave-in, mobile equipment, underground utilities, slips/trips	Daily excavation inspection, sloping/protective system, spotter, locates verified, barricades
Directional bore under roadway	Rotating equipment, pinch points, hydraulic pressure, live traffic	Machine guarding, two-person operation, cones/flagger, hands clear of rotating rod
Loading spoils / haul-off	Overhead loads, mobile equipment, dust	Exclusion zone, spotter, backup alarms verified, dust suppression, hard hats/hi-vis
Potholing / hand digging near gas main	Line strike, flammable atmosphere	Hand dig within 24 in of marks, gas monitor, no spark-producing tools, utility standby
Traffic control setup / teardown	Live traffic, struck-by	TTC per plan, taper per MUTCD, class 3 hi-vis, work behind barrier where possible
Housekeeping / demob	Slips/trips, manual handling	Clear walkways, team lifts over 50 lb, cut-resistant gloves

Specific instructions/items discussed during on-site meeting: Reviewed lane closure timing with DOT inspector; paving crew east of Sta 126 - coordinate truck access; heat advisory after 1 PM - water/shade breaks every hour.
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Energy wheel



Energy type	Examples	
Gravity	- Excavation cave-in - Falling or sliding materials/objects	- Slips/trips/falls - Working at heights
Motion	- Wind - Road/ground conditions - Flying particles/debris - Simultaneous operations	- Watercourses - Ergonomics - Congestion - Vehicles/vessels/mobile equipment
Mechanical	- Tool/equipment nip points/pinch points - Vibration	Rotating equipment
Electrical	- Electrical equipment/lines - normal or abnormal condition (shock or arc flash) - Non-intrinsically safe tools/equipment - Static electricity	
Pressure	- Compressed cylinders - Pressurized piping/hoses/equipment	- Tanks/vessels - Pressure relief systems
Sound	- Tools/equipment - Pressure relief systems	Purging
Radiation	- Welding arc - NDT/X-ray - NORM	- Infrared scanners - Sun
Biological	- Plants - Insects - Needles - Reptiles - Viruses	- Animals - Mold - Bloodborne pathogens - Birds - Bacteria
Chemical	- Flammable/combustible - Toxic vapors/dusts/fibers/fumes - Designated substances, pipeline contaminants, spills, suspect soils - Reactive	- Corrosive - Skin/eye irritants
Temperature	- Cold surfaces (Nitrogen, LNG, propane) - Hot surfaces (friction, heat sources) - Hot emissions/vapors	- Weather conditions - Ignition sources
Human factors	- Knowledge/skill - Risk tolerance - Working alone - Training	- Communication - Fit for duty - Deviation from plan

Have all known hazards been identified and addressed using the energy wheel? ☒ Yes ☐ No

(Do not proceed until you can answer "yes")

Worker/visitor sign off

All workers/visitors must review and sign this form prior to commencing work or upon arrival to the site and repeat process if there are any changes to tasks or site conditions.

Worker/visitor: I have reviewed all applicable documentation, site hazards, and my responsibilities to follow safe work plans to protect myself and others while on site.

Worker/visit name	Worker/visitor signature	Worker/visit name	Worker/visitor signature
Miguel Rodriguez		Jose Martinez	
Juan Lopez		Roberto Diaz	
David Garcia		Luis Hernandez	
Chris Boone		Sam Nguyen	
Terrell Whitaker		Alek Petrov	